

## TITU, Romania

WMO: 154190

Lat: 44.6528N

Lon: 25.5792E

Elev: 526

StdP: 14.42

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 9.4        | 14.4 | 6.3                           | 7.7 | 11.1 | 10.7 | 9.7 | 16.0 | 32.4                  | 33.6 | 22.9 | 29.8 | 2.7                   | 230  | 0.447 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 23.2                   | 92.2            | 70.6 | 89.7 | 70.0 | 87.1 | 69.2 | 73.5                | 87.5 | 72.0 | 85.4 | 70.8 | 83.6 | 3.6                  | 70   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.2                            | 109.6 | 78.6 | 67.9 | 104.6 | 76.9 | 66.7 | 100.1 | 75.7 | 37.4          | 87.9 | 36.0 | 85.2 | 35.0 | 83.8 | 79.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |
|                   |      |      | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |
| 21.0              | 16.4 | 13.4 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |
|                   |      |      | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |
|                   |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 52.7   | 28.2  | 35.6  | 43.6  | 52.1  | 62.9  | 70.8  | 72.9  | 72.9  | 63.8  | 52.8  | 41.6  | 34.0 |
|                                                                               | DBStd       | 16.81  | 8.36  | 8.33  | 8.02  | 7.74  | 5.56  | 5.10  | 4.89  | 5.05  | 6.60  | 7.01  | 7.73  | 7.07 |
|                                                                               | HDD50       | 2192   | 675   | 407   | 225   | 67    | 1     | 0     | 0     | 0     | 0     | 53    | 268   | 496  |
|                                                                               | HDD65       | 5283   | 1140  | 824   | 663   | 388   | 104   | 10    | 3     | 5     | 102   | 382   | 701   | 961  |
|                                                                               | CDD50       | 3173   | 0     | 2     | 27    | 130   | 401   | 623   | 710   | 709   | 415   | 139   | 17    | 0    |
|                                                                               | CDD65       | 792    | 0     | 0     | 0     | 2     | 39    | 184   | 248   | 250   | 67    | 2     | 0     | 0    |
|                                                                               | CDH74       | 8221   | 0     | 0     | 3     | 86    | 474   | 1760  | 2495  | 2609  | 733   | 61    | 0     | 0    |
|                                                                               | CDH80       | 3200   | 0     | 0     | 0     | 9     | 101   | 648   | 1038  | 1170  | 226   | 8     | 0     | 0    |
| Wind                                                                          | WSAvg       | 4.4    | 4.9   | 5.3   | 5.8   | 5.7   | 5.1   | 3.6   | 3.1   | 3.1   | 3.8   | 3.8   | 4.1   | 4.9  |
| Precipitation                                                                 | PrecAvg     | 24.6   | 1.5   | 1.1   | 1.5   | 2.1   | 2.8   | 3.1   | 2.9   | 1.8   | 2.3   | 2.0   | 1.5   | 1.8  |
|                                                                               | PrecMax     | 37.8   | 3.0   | 3.1   | 3.1   | 5.9   | 6.7   | 7.6   | 7.4   | 7.6   | 4.8   | 4.1   | 3.9   | 5.7  |
|                                                                               | PrecMin     | 13.5   | 0.1   | 0.0   | 0.0   | 0.7   | 0.5   | 1.1   | 1.0   | 0.1   | 0.3   | 0.0   | 0.1   | 0.0  |
|                                                                               | PrecStd     | 5.7    | 0.8   | 0.7   | 1.0   | 1.3   | 1.6   | 1.5   | 1.7   | 1.5   | 1.4   | 1.2   | 1.0   | 1.3  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 51.0  | 63.3  | 72.9  | 80.8  | 87.2  | 93.4  | 95.1  | 96.5  | 90.3  | 80.6  | 68.6  | 56.8 |
|                                                                               |             | MCWB   | 44.9  | 52.8  | 56.3  | 61.7  | 65.8  | 72.6  | 72.3  | 72.1  | 67.9  | 62.8  | 57.8  | 48.7 |
|                                                                               | 2%          | DB     | 45.9  | 56.4  | 67.0  | 76.1  | 83.0  | 89.8  | 91.9  | 92.1  | 85.5  | 74.5  | 61.8  | 50.9 |
|                                                                               |             | MCWB   | 41.9  | 48.7  | 52.5  | 60.1  | 65.3  | 71.4  | 71.2  | 69.8  | 65.5  | 60.8  | 54.1  | 46.3 |
|                                                                               | 5%          | DB     | 41.6  | 51.4  | 61.9  | 71.1  | 79.5  | 86.9  | 88.8  | 89.9  | 81.7  | 69.7  | 56.3  | 46.5 |
|                                                                               |             | MCWB   | 39.7  | 45.6  | 50.2  | 57.0  | 63.6  | 70.1  | 70.3  | 69.4  | 64.7  | 58.5  | 51.6  | 43.2 |
|                                                                               | 10%         | DB     | 38.4  | 46.9  | 57.3  | 67.0  | 75.9  | 83.7  | 85.8  | 86.9  | 77.9  | 65.4  | 52.1  | 42.5 |
|                                                                               |             | MCWB   | 36.7  | 43.0  | 48.3  | 55.4  | 62.3  | 69.0  | 69.5  | 68.8  | 63.6  | 56.4  | 48.9  | 40.0 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 45.7  | 54.3  | 57.3  | 63.0  | 68.8  | 75.3  | 75.4  | 75.1  | 71.2  | 64.0  | 58.3  | 50.1 |
|                                                                               |             | MCDB   | 50.2  | 62.0  | 71.0  | 78.2  | 82.9  | 88.8  | 89.9  | 90.8  | 85.6  | 76.6  | 66.0  | 54.0 |
|                                                                               | 2%          | WB     | 42.4  | 49.8  | 53.8  | 60.9  | 67.1  | 72.9  | 73.4  | 72.9  | 67.5  | 61.6  | 55.4  | 47.0 |
|                                                                               |             | MCDB   | 44.8  | 56.0  | 65.0  | 75.4  | 79.7  | 86.5  | 87.2  | 86.9  | 80.8  | 72.6  | 60.4  | 51.0 |
|                                                                               | 5%          | WB     | 39.9  | 46.5  | 51.3  | 58.6  | 65.3  | 71.5  | 72.0  | 71.3  | 66.0  | 59.5  | 52.6  | 43.9 |
|                                                                               |             | MCDB   | 41.8  | 51.1  | 60.9  | 69.3  | 76.6  | 84.6  | 84.7  | 85.4  | 78.0  | 68.2  | 56.1  | 46.9 |
|                                                                               | 10%         | WB     | 37.3  | 43.6  | 48.8  | 56.3  | 63.4  | 70.1  | 70.7  | 70.0  | 64.6  | 57.3  | 49.6  | 40.9 |
|                                                                               |             | MCDB   | 38.4  | 46.7  | 56.4  | 64.8  | 73.7  | 82.2  | 82.6  | 83.8  | 76.7  | 63.8  | 52.5  | 43.0 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 13.5  | 15.7  | 18.8  | 20.4  | 21.1  | 21.8  | 23.2  | 24.5  | 22.5  | 20.2  | 14.0  | 12.3 |
|                                                                               |             | MCDBR  | 17.2  | 23.4  | 28.9  | 29.1  | 27.4  | 26.7  | 27.4  | 29.1  | 28.9  | 28.3  | 22.8  | 19.2 |
|                                                                               |             | MCWBR  | 13.0  | 15.6  | 15.5  | 14.6  | 11.7  | 10.0  | 9.5   | 9.8   | 11.3  | 14.9  | 14.8  | 14.4 |
|                                                                               | 5% WB       | MCDBR  | 15.9  | 22.8  | 26.6  | 26.7  | 24.5  | 25.3  | 24.6  | 26.1  | 25.4  | 26.4  | 20.3  | 20.3 |
|                                                                               |             | MCWBR  | 12.2  | 15.4  | 15.1  | 13.5  | 11.5  | 10.2  | 9.8   | 9.6   | 10.5  | 14.4  | 13.3  | 14.7 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.336  | 0.368 | 0.403 | 0.451 | 0.428 | 0.436 | 0.446 | 0.462 | 0.422 | 0.393 | 0.367 | 0.340 |      |
|                                                                               | taud        | 2.299  | 2.219 | 2.179 | 2.070 | 2.212 | 2.226 | 2.207 | 2.153 | 2.231 | 2.291 | 2.311 | 2.317 |      |
|                                                                               | Ebn at Noon | 241    | 253   | 260   | 257   | 268   | 266   | 261   | 251   | 252   | 242   | 227   | 226   |      |
|                                                                               | Edh at Noon | 26     | 34    | 41    | 49    | 44    | 44    | 44    | 45    | 38    | 32    | 26    | 24    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 442    | 704   | 1090  | 1477  | 1826  | 1983  | 2008  | 1787  | 1298  | 850   | 472   | 362   |      |
|                                                                               | RadStd      | 64     | 105   | 117   | 151   | 144   | 143   | 135   | 110   | 103   | 105   | 59    | 55    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (69 neighbors) | +0.88 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | -256        | +113  | N/A   |       |

Nomenclature: See separate page